ITEC634 – Week 4 Lab 2 Report

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Week 4 Lab 2 – ASP.NET Web Form Application

1. Introduction

This lab focused on developing a simple web application using ASP.NET Razor Pages. The objective was to create a single-page web interface that allows users to input their name and select their favorite programming language, and then display the result dynamically on the same page after submission. The lab provided practical exposure to handling user input, form submission using HTTP POST, and integrating frontend and backend logic within an ASP.NET application.

2. Objectives of the Lab

The main objectives of this lab were:

- To understand how to build a basic ASP.NET web application.
- To implement user input using form controls (text input and dropdown).
- To handle form submission using HTTP POST requests.
- To process user input on the server side using C#.
- To dynamically display results on the same web page.

3. Implementation Process

Project Setup

The application was developed using Visual Studio Code and the .NET SDK. A new ASP.NET Razor Pages project was created using the `dotnet new webapp` command. This generated the required project structure, including the Pages folder containing the UI and backend files.

User Interface Design

The user interface was implemented in the `Index.cshtml` file. It included:

- A text input field for entering the user's name
- A dropdown list with three options: C#, Java, and Python
- A submit button

The form was configured to use the POST method to send data to the server.

Backend Logic

The backend logic was implemented in `Index.cshtml.cs`. Properties were created using the `[BindProperty]` attribute to capture user input from the form.

When the user clicks the submit button, the `OnPost()` method is triggered. This method processes the input and generates a message displaying the user's name and selected programming language.

Displaying Results

The result was displayed dynamically on the same page using Razor syntax. The message was shown below the form after submission, ensuring that the application met the requirement of a single-page interaction.

4. Key Features of the Application

- Accepts user input through a text field
- Provides a dropdown list with predefined options
- Uses HTTP POST for form submission
- Displays output dynamically without navigating to another page
- Maintains a simple and user-friendly interface

5. Challenges Encountered and Solutions

During the development process, a few challenges were encountered:

- Property Binding Errors

Initially, errors occurred because the `Message` property was not properly defined in the backend. This was resolved by correctly declaring the property in the `IndexModel` class.

- File Lock Error

An error occurred where the application executable file was locked during build. This was caused by a previously running instance of the application. The issue was resolved by stopping the running process using `CTRL + C` or terminating the process using `taskkill`.

- Nullability Warnings

Warnings related to non-nullable properties were encountered. These were resolved by making the properties nullable using `string?`.

6. What I Learned

This lab significantly improved my understanding of ASP.NET web development. I learned how frontend and backend components interact within a Razor Pages application. Specifically, I gained knowledge in handling user input, processing form data using HTTP POST requests, and dynamically updating the user interface.

I also developed a better understanding of debugging common issues such as binding errors and file locking problems. Additionally, I became more comfortable using Visual Studio Code and the .NET CLI to build and run web applications.

Overall, this lab provided a strong foundation for building interactive web applications using ASP.NET.

7. Use of Artificial Intelligence (AI)

AI tools such as ChatGPT were used to support the development of this lab. The AI assisted in:

- Understanding ASP.NET project structure
- Debugging errors related to property binding and build issues
- Providing step-by-step guidance for implementation
- Explaining how HTTP POST and Razor Pages work

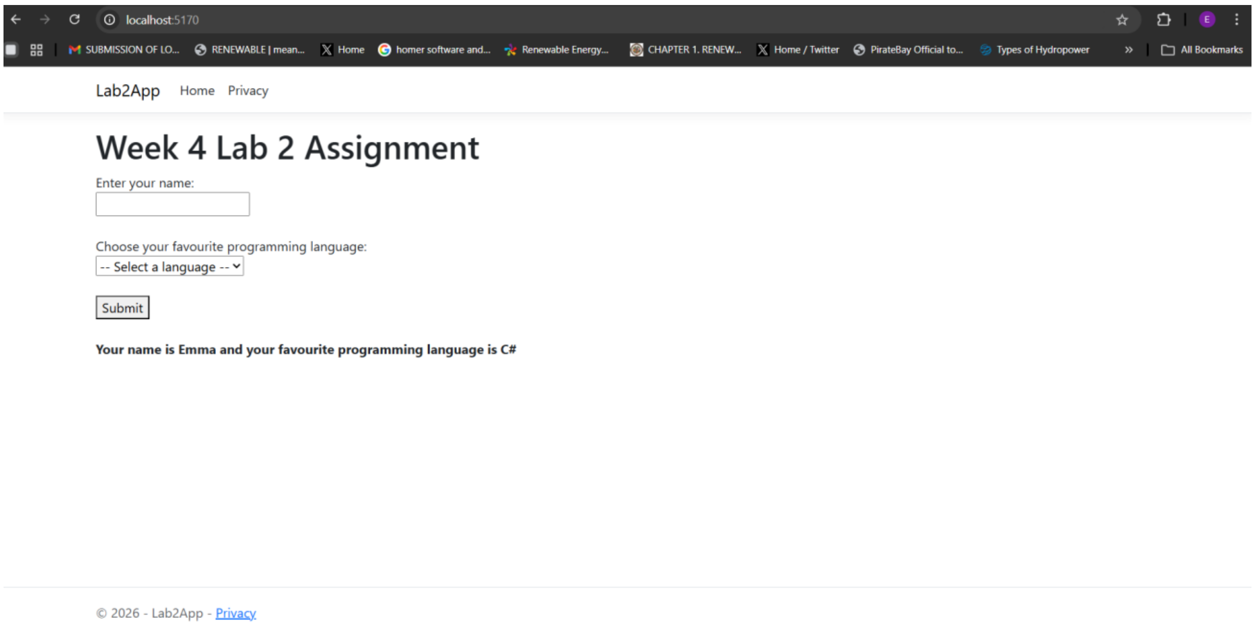
Summary of Prompts Used

- How to create and run an ASP.NET web app in VS Code
- How to implement form submission using POST in Razor Pages

- Debugging errors such as missing properties and file lock issues
- Improving and validating the application logic

Summary of AI Responses

The AI provided structured guidance, code examples, and troubleshooting steps. It helped clarify concepts and resolve errors efficiently, which improved the overall development process and learning experience.



Screenshot of Application

8. Conclusion

In conclusion, this lab successfully demonstrated how to build a simple interactive web application using ASP.NET Razor Pages. The objectives of capturing user input, processing data using POST requests, and displaying results dynamically were achieved.

The lab enhanced my practical skills in web development and provided valuable experience in debugging and problem-solving. This knowledge will be useful for developing more advanced web applications in future coursework and projects.